

# Momentum and impulse

Force over time and velocity changes

## YOUR AIM TODAY

**Use signed momentum and connect impulse to force.**

### YOUR RULE

Read the notes and copy each worked step. Attempt every question before opening page 4. Mark in another colour and repair every error.

Student: Rohan

Start time: \_\_\_\_\_ Finish time: \_\_\_\_\_

Date: \_\_\_\_\_

# The ideas you need

Momentum includes direction, so choose a positive direction before calculating.

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- Momentum  $p=mv$ .
- Impulse=change in momentum= $m(v-u)$ .
- For constant force, impulse= $Ft$ .
- Newton's second law can be written  $F=\text{rate of change of momentum}$ .
- Use negative velocity for motion opposite the chosen positive direction.

## MEMORY RULE

Choose positive, write signs, then use final minus initial.

## Impulse

### WORKED STEPS

3 kg changes from +4 to -2 m/s: impulse= $3(-2-4)=-18$  Ns.

## Average force

### WORKED STEPS

If that happens in 0.5 s,  $F=-18/0.5=-36$  N.

# Independent practice

Show every step. Include units in Mechanics.

| Question                                                   | Working and final answer |
|------------------------------------------------------------|--------------------------|
| 1. Find momentum of 5 kg at 7 m/s.                         |                          |
| 2. Find impulse when 2 kg changes from 3 to 9 m/s.         |                          |
| 3. Find force if change occurs in 0.4 s.                   |                          |
| 4. A 0.2 kg ball changes from +10 to -6 m/s. Find impulse. |                          |
| 5. If contact lasts 0.08 s, find average force.            |                          |
| 6. Explain the meaning of a negative impulse.              |                          |

**STOP**

Do not continue until all six questions have an attempt.

# Mark, diagnose, correct

A wrong answer is useful only when you repair the first wrong step.

| Q | Answer and essential working                                |
|---|-------------------------------------------------------------|
| 1 | 35 kg m/s.                                                  |
| 2 | $2(9-3)=12 \text{ N}\cdot\text{s}$ .                        |
| 3 | $12/0.4=30 \text{ N}$ .                                     |
| 4 | $0.2(-6-10)=-3.2 \text{ N}\cdot\text{s}$ .                  |
| 5 | $-3.2/0.08=-40 \text{ N}$ .                                 |
| 6 | The impulse acts opposite to the chosen positive direction. |

| Q | First wrong step | Correct rule |
|---|------------------|--------------|
|   |                  |              |
|   |                  |              |
|   |                  |              |

## SEND FOR MARKING

Send pages 3 and 4. Ask for a score, the first error in each answer, and one follow-up question per weak skill.